Neuronal Network Activity Across Development and Pharmacological Conditions

Group 35

Abstract*— Human pluripotent stem cell–derived neuronal networks cultured on microelectrode arrays were analyzed to characterize spontaneous activity during development and under pharmacological modulation. Extracellular recordings from DIV 3–35 and drug-treated networks at DIV29 were processed using a unified Python-based pipeline including filtering, spike and burst detection, and network-level metrics. Mean firing rate, burst duration, and spikes per burst were quantified and validated against ground truth data. Pharmacological interventions induced distinct activity profiles, ranging from partial modulation to complete suppression under tetrodotoxin. Relative similarity analysis indicated that certain drug-induced activity patterns partially overlap with early developmental stages; however, these similarities do not reflect equivalent network states. Overall, the results show that activity-based metrics alone are insufficient to equate pharmacologically induced conditions with physiological neuronal development, highlighting the importance of biological context when interpreting network dynamics.

***Keywords*—Burst, spike, firing rate, hPSC, neurons**

I. Introduction

In this project, we analyzed a multi-stage MEA dataset composed of human pluripotent stem cells–derived neuronal networks recorded during spontaneous development (DIV 3–35) and during a pharmacological assay performed at a mature stage (DIV29).

The primary aim of our work was to evaluate the pharmacological treatments administered at DIV29 by directly comparing their effects with the spontaneous activity observed across drug-free developmental days. By characterizing the maturation of firing patterns and burst structure, the natural developmental trajectory served as a benchmark to interpret drug-induced states.

To support this objective, we extracted quantitative features (mean firing rate, burst duration, spikes per burst) using a Python pipeline, compared the DIV29 vehicle baseline with drug-free recordings to distinguish manipulation-induced from naturally occurring states, and assessed whether each treatment shifted the network toward activity regimes resembling earlier or later DIVs.

II. Materials and Methods

A. Cell model and culture conditions

Human pluripotent stem cells–derived cortical neurons (hPSC-derived neurons) were used as the experimental model. These cultures reproduce a mixed glutamatergic/GABAergic cortical phenotype and exhibit long-term spontaneous activity when grown on microelectrode arrays (MEAs) [1]. Neurons were plated on poly-ethylene-imine–coated MEA plates, using 12-well plates for drug-free developmental recordings and 48-well plates for pharmacological experiments. Cultures were maintained at 37 °C and 5% CO_2 throughout the entire experimental period.

Drug-free recordings represent spontaneous network development from DIV3 to DIV35. Pharmacological experiments were performed exclusively at DIV29, a mature stage characterized by stable bursting and synchronized network activity. Importantly, pharmacological manipulations modulate synaptic transmission without altering cell identity or developmental state, allowing direct comparison between natural maturation and drug-induced activity patterns.

B. Pharmacological manipulations

At DIV29, pharmacological agents targeting excitatory and inhibitory synaptic transmission were applied, including:
GABA - “inhibitory agonist” (10 µM),
Gabazine - “GABA_A receptor antagonist” (30 µM),
CNQX - “AMPA/kainate receptor antagonist” (50 µM),
D-AP5 - “NMDA receptor antagonist” (50 µM),
Kainic acid (KA) - “glutamatergic agonist” (1–5 µM), and
Tetrodotoxin (TTX) - “voltage-gated sodium channel blocker” (1 µM).

Moreover, there were also other two different well types: the control wells, which received the vehicle solution without active compounds, and the no-treatment ones, which represented a natural development of the population.

C. MEA recordings

Extracellular signals were acquired using Axion BioSystems Maestro amplifiers at a sampling rate of 12.5

kHz. Drug-free recordings were obtained from 12-well MEAs containing 64 electrodes per well, whereas pharmacological recordings were performed on 48-well MEAs with 16 electrodes per well. All recordings lasted 20 seconds.

D. Data analysis pipeline

All analyses were performed in Python using NumPy, SciPy, pandas, Matplotlib, Seaborn, and standard I/O libraries. The processing pipeline consisted of: (i) data loading, (ii) signal filtering, (iii) spike detection, (iv) validation against ground truth data, (v) burst detection, (vi) extraction of burst-related metrics, and (vii) statistical and graphical comparison across conditions. Statistical metrics were aggregated at the well level and subsequently summarized per DIV or pharmacological condition to obtain network-level descriptors.

E. Signal preprocessing

Raw extracellular signals were preprocessed using a fourth-order Butterworth band-pass filter (300–3000 Hz) to cancel Local Field Potentials. To further reduce common-mode noise, signals were median-demeaned across channels and processed with common median referencing (CMR). This preprocessing pipeline was applied consistently to both drug-free and pharmacological datasets.

F. Spike detection

Spike detection was based exclusively on negative extracellular peaks, as the negative phase typically exhibits larger amplitude and lower variability than the positive component, reducing ambiguity and false detections. Noise levels were estimated using the median absolute deviation (MAD), and detection thresholds were defined as a multiple of the estimated noise of each well (σ). We used F1-Score to decide whether our spike detection algorithm was reliable (more precisely, an F1-Score higher than 75% indicated a good spike sorting technique).

After iterative parameter optimization, three thresholds were selected: 4.3σ for DIV3, 4.7σ for DIV35, and 4.4σ for all remaining DIVs. The lower threshold at DIV3 compensates for the small-amplitude spikes of immature networks, improving sensitivity, whereas the higher threshold at DIV35 preserves specificity in sparsely active mature networks. A refractory period of 1.5 ms was imposed to prevent multiple detections of the same event.

In table 1 are displayed the F1-Score of the spike sorting results of each DIV.

DIV	F1-Score
DIV 3	0.7622
DIV 7	0.7829
DIV 14	0.8309
DIV 21	0.8662
DIV 28	0.8130
DIV 35	0.7486

Table 1. Spike sorting F1-Score metric of natural development DIVs.

In the case of pharmacological wells, we chose 4.4σ as threshold, obtaining these F1-Scores:

DIV 29 Pharma	F1-Score
GABA, GABAZINE, Kainic Acid, CNQX, D-AP5	0.843
TTX	0.7341

Table 2. Spike sorting F1-Score metric of pharmacological plates.

G. Burst detection and burst metrics

Bursts were identified using an interspike interval (ISI)–based algorithm. Bursts were defined as sequences of at least five spikes separated by ISIs shorter than 100 ms, with a minimum burst duration of 100 ms. Burst metrics included mean burst duration, mean spikes per burst, and firing rate. To avoid bias from silent electrodes, burst statistics were computed only from channels exhibiting at least one burst and then averaged at the well level.

H. Relative similarity matrices

Relative similarity matrices were computed to quantitatively compare developmental network activity with pharmacologically induced states. For each metric, median values were calculated per developmental stage (DIV) and per pharmacological condition across wells. Pairwise comparisons between all DIVs and drug conditions were performed using a normalized relative similarity index based on the absolute difference between metric values. Similarity values ranged from 0 to 1, with higher values indicating closer correspondence between developmental and pharmacological profiles.

III. Results

A. Evolution of neuronal activity

In figure 1, it can be seen the spontaneous, drug-free recordings revealing a progressive increase in neuronal activity from DIV3 to DIV28, consistently reflected across all three analyzed metrics: firing rate, mean burst duration, and spikes per burst. This gradual increase is indicative of the progressive maturation of synaptic connectivity and the temporal organization of network activity. At DIV35, a sharp decline in activity is observed.

On the other hand, pharmacological manipulations applied at DIV29 induce pronounced changes in network behavior. In general, treatments with inhibitory effects (GABA, CNQX, D-AP5) reduce firing rate and burst structure, whereas excitatory or disinhibitory (Gabazine and Kainic acid) treatments significantly increase neuronal activity.

Additionally, TTX represents an extreme case: blockade of voltage-gated sodium channels results in an almost complete suppression of spiking activity, leading to the absence of detectable bursts. Due to this fact, bursts-related metrics (mean duration per burst and spikes per burst), regarding this pharmacological treatment, are absent.

The inhibiting treatment caused by CNQX, D-AP5 and GABA can be better seen in the burst structure, as the duration of the burst and the number of spikes in it are close to 0.

Another feature that can be highlighted in the chart, is the behavior of the control wells. It can be seen that by only adding the vehicle to the neuron culture, it causes a preponderant stimulation in the population activity. Indeed, it is the closest boxplot to the Gabazine representation.

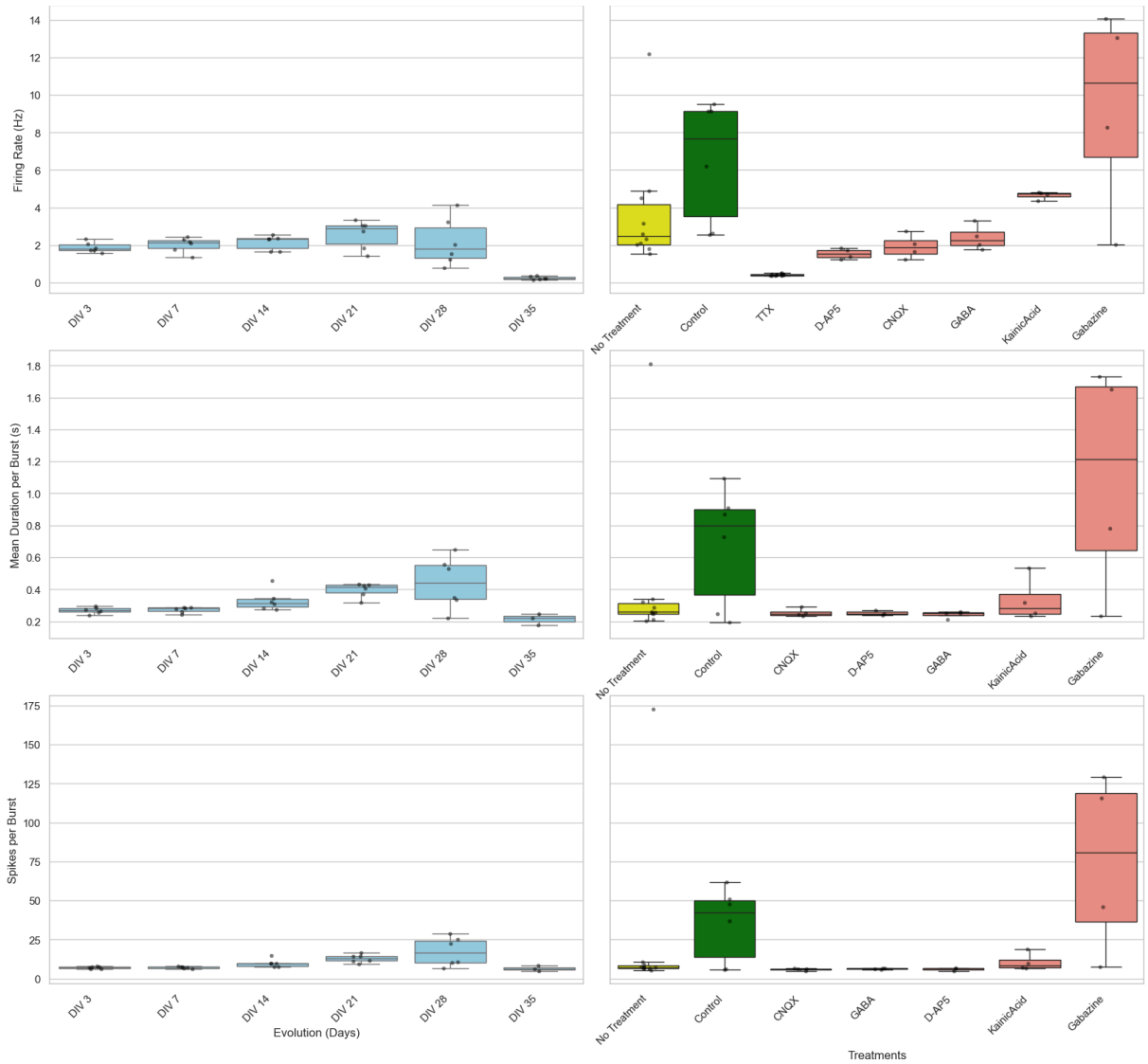

Fig. 1. Natural development vs Pharmacological intervention.

B. Mean firing rate comparison

In figure 2, it is shown the profile comparison regarding the mean firing rate between the natural developments and the pharmacological ones. The values range from 0 to 1, where higher values indicate greater similarity. From our point of view, values above 0.9 represent a very high degree of similarity, indicating that the network exhibits nearly identical firing dynamics

under those conditions. High similarity values are mainly observed at early and mid developmental stages, particularly for CNQX at DIV 3 and DIV 28, and for GABA at DIV 7 and DIV 14, suggesting that these stages share closely comparable firing rate patterns. In contrast, Gabazine, Kainic Acid, and TTX show consistently low similarity values across most developmental stages, indicating firing rate profiles that differ substantially from the natural reference.

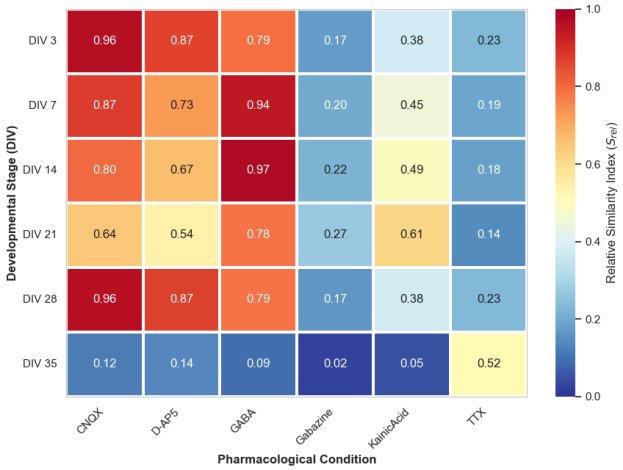

Fig. 2. Profile Comparison: Mean Firing Rate (MFR).

C. *Mean burst duration comparison*

Figure 3 presents the comparison between developmental stages and pharmacological conditions based on mean burst duration. The analysis shows that early developmental stages (DIV 3 and DIV 7) display strong correspondence with CNQX, D-AP5, GABA, and KA, indicating that these pharmacological conditions reproduce burst duration patterns characteristic of immature networks. KA also exhibits a high correspondence at DIV 14, suggesting partial preservation of burst temporal structure at intermediate maturation stages. Moreover, there is a remarkable similarity between DIV35 and CNQX, D-AP5 and GABA. In contrast, Gabazine consistently shows low correspondence across all DIVs. Notably, TTX displays null similarity values at all developmental stages, as the suppression of action potential generation prevents the emergence of bursting activity, resulting in the absence of well-defined burst duration metrics. Overall, these results indicate that burst duration is selectively preserved under specific synaptic modulations, whereas strong network excitation leads to pronounced deviations from the physiological developmental trajectory.

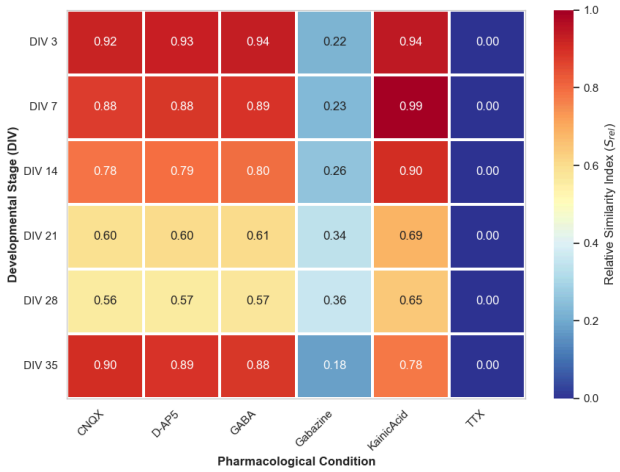

Fig. 3. Profile Comparison: Mean Burst Duration (MBD)

D. *Mean spikes per burst comparison*

In figure 4, the analysis focuses on Mean Spikes per Burst (MSB). High similarity values are observed at early developmental stages under D-AP5 and GABA, with values above 0.9 at DIV 3 and DIV 7, indicating closely comparable burst spike content during early network maturation. In the case of CNQX, it shows high similarity at DIV 35, suggesting that the spike composition of bursts in mature networks closely resembles the reference profile under AMPA receptor blockade. As development progresses toward intermediate stages (DIV 14 to DIV 28), similarity values decrease across most pharmacological conditions, reflecting a progressive divergence in the number of spikes contained within bursts as network dynamics become more complex. KA maintains moderate to high similarity across early and mid stages. Gabazine consistently shows low similarity values, highlighting markedly altered burst spike composition under disinhibition. As mentioned before, TTX shows zero similarity across all developmental stages, as action potential generation is widely suppressed, preventing the formation of bursts and making the mean number of spikes per burst undefined.

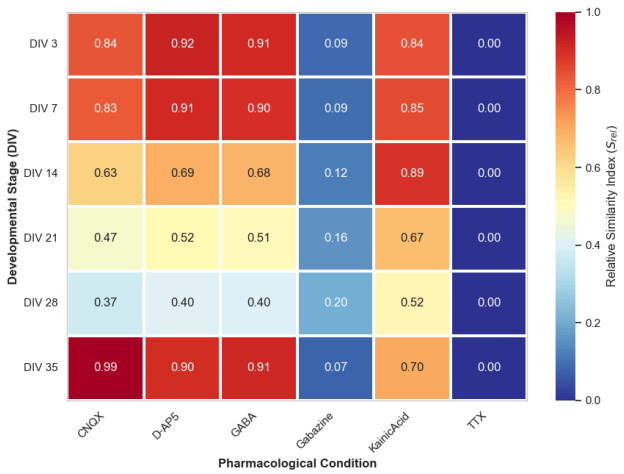

Fig. 4. Profile Comparison: Mean Spikes per Burst (MSB)

IV. Discussion

To address the original research question, this study explored whether pharmacologically induced activity states in mature neuronal networks (DIV29) could be meaningfully interpreted in relation to spontaneous activity patterns observed during drug-free development.

Pharmacological interventions applied at DIV29 produced marked changes in network activity, consistent in direction with their expected functional outcomes at the network level. Inhibitory treatments generally reduced firing rate and burst-related metrics, whereas excitatory or disinhibitory conditions increased overall activity. Importantly, the present work does not rely on mechanistic or receptor-level interpretations of these effects. Instead, conclusions are drawn strictly from observed changes in network-level activity patterns, without attributing them to specific physiological pathways.

A central observation of this study is that numerical similarity between pharmacological conditions and early developmental stages does not necessarily imply functional equivalence. In particular, although treatment with CNQX at DIV29 reduces firing rate and burst metrics to values comparable to those observed at DIV3, this correspondence is purely numerical and lacks functional significance at the network level. DIV3 reflects an immature network state in which low activity arises from incomplete connectivity and ongoing organizational processes, whereas CNQX induces a pharmacologically constrained state in an already mature network. Consequently, the resulting activity pattern does not correspond to any natural stage of neuronal development[3][4].

This consideration extends to other pharmacological treatments that directly perturb specific components of synaptic transmission. While such interventions may yield low values in certain metrics and thus appear similar to early developmental stages, these states represent experimentally induced configurations rather than genuine developmental equivalents. Without careful interpretation, such similarities could lead to misleading conclusions.

Several limitations of this study must be acknowledged. Pharmacological experiments were conducted at a single developmental time point, restricting the ability to assess how drug effects interact with ongoing maturation. The elevated activity occasionally observed in vehicle control conditions further contributes to uncertainty, although DIV29 remained the only available reference point for pharmacological comparisons. Finally, technical limitations associated with spike detection in low-activity regimes, particularly under strong inhibitory conditions and at late developmental stages, introduce additional sources of variability.

This study demonstrates that while pharmacological manipulation of mature neuronal networks can produce activity levels that numerically resemble those observed during early development, such similarities are largely context-dependent and do not necessarily reflect equivalent functional states.

Overall, these findings highlight the importance of cautious interpretation when comparing developmental and pharmacologically induced network states and underscore the need for future studies incorporating larger multi-stage pharmacological interventions.

In conclusion, neuronal networks cannot be meaningfully assimilated or classified solely on the basis of the three activity features analyzed in this study. While these metrics capture quantitative aspects of network activity, they are insufficient to establish functional equivalence across developmental and pharmacologically induced states. A more accurate interpretation requires the incorporation of additional factors that account for the biological context of the network, particularly its maturation state and underlying organizational dynamics.

References

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